Adjoint kernels for a self-gravitating, hydrostatic earth model from a weak formulation

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Abstract

We derive sensitivity kernels for a non-rotating, self-gravitating earth model in hydrostatic equilibrium, for perturbations in the elastic tensor (including its anelastic dispersion), the density, and the source, with respect to one-dimensional perturbations and three-dimensional perturbations of a one-dimensional background. The derivation starts from the weak form of the equations of motion, in which the boundary conditions are natural, and treats the equilibrium gravitational potential Φ as an independent variable constrained by Poisson's equation. The effect of a density perturbation on the equilibrium gravity is then carried by a single, static adjoint potential λ, which satisfies a Poisson equation whose source is formed, once, from the forward and adjoint fields summed over frequency. Every kernel is then local: there is no need to substitute an explicit expression for the perturbation $\delta\Phi$ and exchange the order of integration, as in Liu & Tromp (2008) and Eaton et al. (2026), and the result agrees with the latter. The method is formulated in the frequency domain at complex frequencies, as in Adourian et al. (2024), with the adjoint weights derived from the same discrete transformation that forms the synthetic seismograms, so that the kernels predict finite differences of the computed misfit exactly. Topography on discontinuities is outlined in Section 8.

1 Setting and notation

The earth model occupies a ball V of radius a, divided into regions by internal discontinuities $\Sigma = \Sigma_{SS} \cup \Sigma_{FS}$ (solid–solid and fluid–solid), with free surface ∂V. The unit normal $\hat{\mathbf{n}}$ points from the $-$ side of a boundary to the $+$ side, and $[f]^+_- = f_+ - f_-$; on ∂V the $+$ side is outside. With this convention, for a tensor field $\mathbf{v}$ smooth in each region,

$$\int_V \nabla \cdot \mathbf{v}\, \mathrm{d}V = -\int_{\Sigma \cup \partial V} [\hat{\mathbf{n}} \cdot \mathbf{v}]^+_-\, \mathrm{d}\Sigma, \tag{1}$$

with $\mathbf{v} = 0$ outside V. For vector fields we use the bilinear pairing $\langle \mathbf{a}, \mathbf{b} \rangle = \int_V \mathbf{a} \cdot \mathbf{b}\, \mathrm{d}V$, without complex conjugation; all forms below are bilinear, and the frequency-domain operators are complex symmetric rather than Hermitian.

The equilibrium model has density ρ, elastic tensor $\mathbf{c}$ (the incremental tensor of the pre-stressed state, with the major and minor symmetries) and gravitational potential Φ, with

$$\nabla^2 \Phi = 4\pi G \rho \quad \text{in } \mathbb{R}^3, \qquad \Phi \to 0 \text{ as } |\mathbf{x}| \to \infty. \tag{2}$$

It is in hydrostatic equilibrium: the initial stress is a pressure p_0 with $\nabla p_0 = -\rho \nabla \Phi$, so that $\nabla \rho \parallel \nabla \Phi$ in each region and the boundaries are level surfaces of Φ. For a spherically symmetric model, $\nabla \Phi = g \hat{\mathbf{r}}$ with $g = 4\pi G r^{-2} \int_0^r \rho r'^2\, \mathrm{d}r' \geq 0$.

2 The forward problem

2.1 The hydrostatic equations

In the hydrostatic form (Dahlen & Tromp, 1998, eq. 3.259), the initial pressure has been eliminated with $\nabla p_0 = -\rho\nabla\Phi$, and the displacement $\mathbf{s}$ and the perturbation ϕ of the gravitational potential satisfy, without rotation,

$$\rho\ddot{\mathbf{s}} - \nabla\cdot\mathbf{T} + \nabla(\rho\mathbf{s}\cdot\nabla\Phi) - \nabla\cdot(\rho\mathbf{s})\nabla\Phi + \rho\nabla\phi = \mathbf{f}, \qquad \mathbf{T} = \mathbf{c} : \nabla\mathbf{s}, \tag{3}$$

$$(4\pi G)^{-1}\nabla^2\phi + \nabla\cdot(\rho\mathbf{s}) = 0, \tag{4}$$

the first in each region of V, the second in $\mathbb{R}^3$. The boundary conditions are $[\hat{\mathbf{n}}\cdot\mathbf{T}]^+_- = \mathbf{0}$ on $\Sigma\cup\partial V$ (with $\mathbf{T}=0$ outside), $[\mathbf{s}]^+_- = \mathbf{0}$ on Σ_{SS}, $[\hat{\mathbf{n}}\cdot\mathbf{s}]^+_- = 0$ and $\hat{\mathbf{n}}\cdot\mathbf{T} \parallel \hat{\mathbf{n}}$ on Σ_{FS}, and $[\phi]^+_- = 0$, $[\hat{\mathbf{n}}\cdot\boldsymbol{\xi}]^+_- = 0$ with $\boldsymbol{\xi} = (4\pi G)^{-1}\nabla\phi + \rho\mathbf{s}$ on every boundary, with $\phi\to 0$ at infinity (Liu & Tromp, 2008). These are the equations that YSpec solves, written in terms of the radial variables of Section 9.

Integrating (3) through a thin layer across a boundary, the distributional parts of $\nabla(\rho\mathbf{s}\cdot\nabla\Phi)$ and $-\nabla\cdot(\rho\mathbf{s})\nabla\Phi$ are $\hat{\mathbf{n}}\,[\rho]^+_-\, g\,\hat{\mathbf{n}}\cdot\mathbf{s}$ and $-\,[\rho]^+_-\,(\hat{\mathbf{n}}\cdot\mathbf{s})g\,\hat{\mathbf{n}}$, which cancel because $\nabla\Phi = g\hat{\mathbf{n}}$ there and $\hat{\mathbf{n}}\cdot\mathbf{s}$ is continuous. The equations therefore hold in each region with only the stated conditions on the boundaries.

2.2 The weak form

Let $\mathcal{U}$ be the displacements that honour the kinematic conditions ($\mathbf{s}$ continuous on Σ_{SS}, $\hat{\mathbf{n}}\cdot\mathbf{s}$ continuous on Σ_{FS}) and $\mathcal{P}$ the potentials in $H^1(\mathbb{R}^3)$ that decay at infinity. Take $\mathbf{s}'\in\mathcal{U}$, multiply (3) by it, integrate over V, and use (1):

- $-\langle\mathbf{s}', \nabla\cdot\mathbf{T}\rangle = \int_V \nabla\mathbf{s}' : \mathbf{T}\,\mathrm{d}V + \int [\mathbf{s}'\cdot\hat{\mathbf{n}}\cdot\mathbf{T}]^+_-\,\mathrm{d}\Sigma$, and the surface term vanishes: on Σ_{SS} both $\mathbf{s}'$ and $\hat{\mathbf{n}}\cdot\mathbf{T}$ are continuous, on Σ_{FS} the traction is $\tau\hat{\mathbf{n}}$ with τ and $\hat{\mathbf{n}}\cdot\mathbf{s}'$ continuous, and on ∂V the traction is zero.
- $\langle\mathbf{s}', \nabla(\rho\mathbf{s}\cdot\nabla\Phi)\rangle - \langle\mathbf{s}', \nabla\cdot(\rho\mathbf{s})\nabla\Phi\rangle = \int_V \rho\,[\mathbf{s}\cdot\nabla(\mathbf{s}'\cdot\nabla\Phi) - (\nabla\cdot\mathbf{s}')(\mathbf{s}\cdot\nabla\Phi)]\,\mathrm{d}V$ plus the surface terms $\int [-(\hat{\mathbf{n}}\cdot\mathbf{s}')\rho\,\mathbf{s}\cdot\nabla\Phi + (\mathbf{s}'\cdot\nabla\Phi)\rho\,\hat{\mathbf{n}}\cdot\mathbf{s}]^+_-\,\mathrm{d}\Sigma$, which vanish because $\nabla\Phi = g\hat{\mathbf{n}}$ on the boundaries.

Similarly, multiplying (4) by $\phi'\in\mathcal{P}$ and integrating over $\mathbb{R}^3$, the surface terms vanish by the continuity of $\hat{\mathbf{n}}\cdot\boldsymbol{\xi}$. With

$$A_g(\mathbf{s}', \mathbf{s}) = \int_V \rho\left[\mathbf{s}\cdot\nabla\nabla\Phi\cdot\mathbf{s}' + (\mathbf{s}\cdot\nabla\mathbf{s}')\cdot\nabla\Phi - (\nabla\cdot\mathbf{s}')(\mathbf{s}\cdot\nabla\Phi)\right]\,\mathrm{d}V, \tag{5}$$

$$\begin{aligned} B\big((\mathbf{s}', \phi'), (\mathbf{s}, \phi)\big) = &\int_V \nabla\mathbf{s}' : \mathbf{c} : \nabla\mathbf{s}\,\mathrm{d}V + A_g(\mathbf{s}', \mathbf{s}) \\ &+ \int_V \rho\,(\mathbf{s}'\cdot\nabla\phi + \mathbf{s}\cdot\nabla\phi')\,\mathrm{d}V + \frac{1}{4\pi G}\int_{\mathbb{R}^3} \nabla\phi'\cdot\nabla\phi\,\mathrm{d}V, \end{aligned} \tag{6}$$

where $(\mathbf{s}\cdot\nabla\mathbf{s}')\cdot\nabla\Phi = s_i\,\partial_i s'_j\,\partial_j\Phi$, the forward problem is to find $(\mathbf{s}, \phi)$ in $\mathcal{U}\times\mathcal{P}$ such that

$$\langle\mathbf{s}', \rho\ddot{\mathbf{s}}\rangle + B\big((\mathbf{s}', \phi'), (\mathbf{s}, \phi)\big) = \langle\mathbf{s}', \mathbf{f}\rangle \quad \text{for all } (\mathbf{s}', \phi')\in\mathcal{U}\times\mathcal{P}. \tag{7}$$

The equation with $\phi' = 0$ is the weak form of (3); that with $\mathbf{s}' = 0$ is minus that of (4). All the boundary conditions on tractions and fluxes are natural.

Lemma 1. *For a hydrostatic background, A_g, and so B, is symmetric.*

Proof. With $\mathbf{N} = \rho(\mathbf{s} \otimes \mathbf{s}' - \mathbf{s}' \otimes \mathbf{s})$,

$$A_g(\mathbf{s}', \mathbf{s}) - A_g(\mathbf{s}, \mathbf{s}') = \int_V \left[\partial_i N_{ij}\, \partial_j \Phi - (\partial_i \rho)(s_i s'_j - s'_i s_j)\partial_j \Phi\right] \mathrm{d}V.$$

The second term vanishes because $\nabla\rho \parallel \nabla\Phi$. In the first, integrating by parts leaves $-\int N_{ij}\partial_i\partial_j\Phi$, which vanishes because $\mathbf{N}$ is antisymmetric, and the surface term $\int [n_i N_{ij}\partial_j\Phi]_-^+ \,\mathrm{d}\Sigma$, which is $\int [g\,\hat{\mathbf{n}}\cdot\mathbf{N}\cdot\hat{\mathbf{n}}]_-^+ \,\mathrm{d}\Sigma = 0$. □

The symmetry gives reciprocity, and with it the identification of the adjoint problem with a forward one (Section 4). It uses the hydrostatic conditions; we return to this in Section 7.

2.3 Complex frequencies

As in YSpec, the forward problem is solved at the frequencies $\omega_k = 2\pi f_k - i\varepsilon$, $f_k = k\,\Delta f$, $k = k_0, \dots, k_1$, with ε chosen so that the response decays by a factor e^{-5} over the length of the record, and the time series is formed by an inverse transform followed by multiplication by $e^{\varepsilon t}$ (Adourian et al., 2024). The transform of (7), with time dependence $e^{i\omega t}$, is

$$S_k\big((\mathbf{s}', \phi'), (\mathbf{s}_k, \phi_k)\big) := -\omega_k^2 \langle \mathbf{s}', \rho \mathbf{s}_k\rangle + B_k\big((\mathbf{s}', \phi'), (\mathbf{s}_k, \phi_k)\big) = \langle \mathbf{s}', \mathbf{f}_k\rangle, \tag{8}$$

where B_k is B with $\mathbf{c}$ replaced by the complex tensor $\mathbf{c}_k = \mathbf{c}(\omega_k)$, which has the same symmetries. Each S_k is a complex-symmetric bilinear form.

3 Observables and adjoint weights

3.1 Synthetic samples

Let $X_k = \hat{\boldsymbol{\nu}} \cdot \mathbf{s}_k(\mathbf{x}_r)$ be the spectrum of the component $\hat{\boldsymbol{\nu}}$ at the receiver $\mathbf{x}_r$. YSpec forms the samples $u_n = u(t_n)$, $t_n = n\Delta t$, as

$$u_n = \Delta f\, e^{\varepsilon t_n} \sum_k 2\,\mathrm{Re}\left[w_k R_k X_k\, e^{2\pi i f_k t_n}\right], \tag{9}$$

with w_k the (real) taper and R_k the instrument response, both at the complex frequency, and any time derivatives of the displacement included in R_k. For a real functional $J(u_0, u_1, \dots)$ of the samples,

$$\delta J = \sum_k 2\,\mathrm{Re}\left[A_k\, \delta X_k\right], \qquad A_k = \Delta f\, w_k R_k \sum_n \frac{\partial J}{\partial u_n}\, e^{\varepsilon t_n} e^{2\pi i f_k t_n}. \tag{10}$$

The adjoint weights A_k are the discrete counterpart of the processed adjoint source of Adourian et al. (2024): the factor $e^{\varepsilon t_n}$ reflects the undamping of the synthetic, and derivatives are taken of the computed samples, so that (10) is exact for the discrete misfit.

3.2 Cross-correlation traveltimes

For the delay τ of a phase in a window $W(t)$, measured by cross-correlation with a reference that is the synthetic itself (Dahlen et al., 2000), a delay $\delta\tau$ changes the synthetic by $\delta u = -\dot{u}\,\delta\tau$, and so

$$\delta\tau = -\frac{\sum_n W_n \dot{u}_n\, \delta u_n}{\sum_n W_n \dot{u}_n^2}, \qquad \frac{\partial\tau}{\partial u_n} = -\frac{W_n \dot{u}_n}{\sum_m W_m \dot{u}_m^2}, \tag{11}$$

with $\dot{u}$ the velocity synthetic. This is the adjoint source of Liu & Tromp (2008, eqs 82–83) in discrete form.

4 The Lagrangian

The model parameters are ρ, $\mathbf{c}$ (through its elastic values and quality factors) and the source. The equilibrium potential depends on ρ through (2); rather than eliminate it, we treat Φ as an independent variable and impose (2) in weak form with a multiplier λ:

$$P(\Phi, \lambda; \rho) := \frac{1}{4\pi G} \int_{\mathbb{R}^3} \nabla\lambda \cdot \nabla\Phi \, \mathrm{d}V + \int_V \rho\lambda \, \mathrm{d}V = 0 \quad \text{for all } \lambda. \tag{12}$$

The Lagrangian is

$$\mathcal{L} = J - \sum_k 2\,\mathrm{Re}\left[S_k\big((\mathbf{s}_k^\circ, \phi_k^\circ), (\mathbf{s}_k, \phi_k)\big) - \langle \mathbf{s}_k^\circ, \mathbf{f}_k \rangle\right] + P(\Phi, \lambda; \rho), \tag{13}$$

with $(\mathbf{s}_k^\circ, \phi_k^\circ)$ the multipliers for the forward problems. Where the forward and equilibrium problems hold, $\mathcal{L} = J$ whatever the multipliers, so $\delta J = \partial_m \mathcal{L}\, \delta m$ once the multipliers make $\mathcal{L}$ stationary with respect to $\mathbf{s}_k$, ϕ_k and Φ.

The adjoint fields. Stationarity with respect to $(\mathbf{s}_k, \phi_k)$, using (10), requires $2\,\mathrm{Re}[A_k \hat{\boldsymbol{\nu}} \cdot \delta\mathbf{s}(\mathbf{x}_r)] = 2\,\mathrm{Re}[S_k((\mathbf{s}_k^\circ, \phi_k^\circ), (\delta\mathbf{s}, \delta\phi))]$ for all complex $(\delta\mathbf{s}, \delta\phi)$, and so, replacing $\delta\mathbf{s}$ by $i\delta\mathbf{s}$ for the imaginary part, and by the symmetry of S_k (Lemma 1),

$$S_k\big((\delta\mathbf{s}, \delta\phi), (\mathbf{s}_k^\circ, \phi_k^\circ)\big) = \langle \delta\mathbf{s}, A_k \hat{\boldsymbol{\nu}}\, \delta(\mathbf{x} - \mathbf{x}_r) \rangle \quad \text{for all } (\delta\mathbf{s}, \delta\phi). \tag{14}$$

The adjoint field at ω_k is the forward response, at the same complex frequency, to the point force $A_k \hat{\boldsymbol{\nu}}$ at the receiver. No time reversal or change of frequency is needed, and the solver is the forward one.

The adjoint potential. Stationarity with respect to Φ requires

$$\frac{1}{4\pi G} \int_{\mathbb{R}^3} \nabla\lambda \cdot \nabla\psi \, \mathrm{d}V = \int_V (\mathbf{M} : \nabla\nabla\psi + \mathbf{m} \cdot \nabla\psi) \, \mathrm{d}V \quad \text{for all } \psi, \tag{15}$$

where, from (5), the forward and adjoint fields enter through the static fields

$$\mathbf{M} = \sum_k 2\,\mathrm{Re}\left[\rho\ \mathrm{sym}(\mathbf{s}_k \otimes \mathbf{s}_k^\circ)\right], \qquad \mathbf{m} = \sum_k 2\,\mathrm{Re}\left[\rho\left((\mathbf{s}_k \cdot \nabla)\mathbf{s}_k^\circ - \mathbf{s}_k\, \nabla \cdot \mathbf{s}_k^\circ\right)\right]. \tag{16}$$

This is a single, static Poisson problem, solved once after the frequency sum (Section 6).

5 The kernels

With the multipliers so chosen,

$$\delta J = -\sum_k 2\,\mathrm{Re}\left[\partial_m S_k\big((\mathbf{s}_k^\circ, \phi_k^\circ), (\mathbf{s}_k, \phi_k)\big)\delta m\right] + \sum_k 2\,\mathrm{Re}\langle \mathbf{s}_k^\circ, \delta\mathbf{f}_k \rangle + \int_V \lambda\, \delta\rho \, \mathrm{d}V. \tag{17}$$

Elastic tensor. $\delta J = \int_V \mathbf{K}_c :: \delta\mathbf{c} \, \mathrm{d}V$ with

$$\mathbf{K}_c = -\sum_k 2\,\mathrm{Re}\left[\frac{\partial \mathbf{c}_k}{\partial \mathbf{c}}^{\mathsf{T}} : \left(\nabla\mathbf{s}_k^\circ \otimes \nabla\mathbf{s}_k\right)\right], \tag{18}$$

the chain rule through the dispersion. For transverse isotropy with a radial axis, in terms of the strains $\boldsymbol{\epsilon} = \mathrm{sym}\,\nabla\mathbf{s}$ and $\boldsymbol{\epsilon}^\circ = \mathrm{sym}\,\nabla\mathbf{s}^\circ$ at frequency ω_k, with h the horizontal components, the elastic energy density is $C\epsilon_{rr}^\circ\epsilon_{rr} + A\,\mathrm{tr}_h\boldsymbol{\epsilon}^\circ\,\mathrm{tr}_h\boldsymbol{\epsilon} + F(\epsilon_{rr}^\circ \mathrm{tr}_h\boldsymbol{\epsilon} + \mathrm{tr}_h\boldsymbol{\epsilon}^\circ\epsilon_{rr}) + 4L\,\boldsymbol{\epsilon}_{rh}^\circ \cdot \boldsymbol{\epsilon}_{rh} + N(2\boldsymbol{\epsilon}_{hh}^\circ :$

$\boldsymbol{\epsilon}_{hh} - 2\,\mathrm{tr}_h\boldsymbol{\epsilon}^\circ\,\mathrm{tr}_h\boldsymbol{\epsilon}$), every term invariant under rotation about $\hat{\mathbf{r}}$, so the two fields may be expressed in different horizontal frames. With D_k the dispersion of YSpec, so that a modulus X with inverse quality factor q_X is $X(1+q_X D_k)$ at ω_k,

$$K_X = -\sum_k 2\,\mathrm{Re}\left[(1+q_X D_k)\,e_X(\boldsymbol{\epsilon}_k^\circ, \boldsymbol{\epsilon}_k)\right], \qquad K_{q_X} = -\sum_k 2\,\mathrm{Re}\left[X D_k\,e_X(\boldsymbol{\epsilon}_k^\circ, \boldsymbol{\epsilon}_k)\right], \tag{19}$$

for $X \in \{A, C, F, L, N\}$ and e_X its coefficient above. For isotropy, $K_\kappa = -\sum_k 2\,\mathrm{Re}[(1+q_\kappa D_k)\,\nabla\cdot\mathbf{s}_k^\circ\,\nabla\cdot\mathbf{s}_k]$ and $K_\mu = -\sum_k 2\,\mathrm{Re}[(1+q_\mu D_k)\,2\mathbf{d}_k^\circ : \mathbf{d}_k]$ with $\mathbf{d}$ the deviatoric strain, as in Liu & Tromp (2008, eqs 63–64). Attenuation kernels come at no extra cost.

Density, at fixed moduli. From (8), (5), (6) and (17),

$$\begin{aligned} K_\rho = \sum_k 2\,\mathrm{Re}\Big[&\omega_k^2\,\mathbf{s}_k^\circ\cdot\mathbf{s}_k - \mathbf{s}_k^\circ\cdot\nabla\phi_k - \mathbf{s}_k\cdot\nabla\phi_k^\circ \\ &- \mathbf{s}_k\cdot\nabla\nabla\Phi\cdot\mathbf{s}_k^\circ - (\mathbf{s}_k\cdot\nabla\mathbf{s}_k^\circ)\cdot\nabla\Phi + (\nabla\cdot\mathbf{s}_k^\circ)(\mathbf{s}_k\cdot\nabla\Phi)\Big] + \lambda. \end{aligned} \tag{20}$$

Every term is local; the equilibrium gravity's dependence on the density is carried entirely by λ. For a spherically symmetric background the gravitational terms simplify to

$$\mathbf{s}\cdot\nabla\nabla\Phi\cdot\mathbf{s}^\circ + (\mathbf{s}\cdot\nabla\mathbf{s}^\circ)\cdot\nabla\Phi - (\nabla\cdot\mathbf{s}^\circ)(\mathbf{s}\cdot\nabla\Phi) = \left(4\pi G\rho - \frac{4g}{r}\right) s_r s_r^\circ + g\left(\mathbf{s}_h\cdot\nabla_h s_r^\circ - s_r\nabla_h\cdot\mathbf{s}_h^\circ\right), \tag{21}$$

using $\nabla\nabla\Phi = \Phi''\hat{\mathbf{r}}\hat{\mathbf{r}} + (g/r)(\mathbf{I} - \hat{\mathbf{r}}\hat{\mathbf{r}})$ with $\Phi'' = 4\pi G\rho - 2g/r$, and $(\mathbf{s}\cdot\nabla)\hat{\mathbf{r}} = \mathbf{s}_h/r$, with ∇_h the horizontal gradient. The coefficient $-4g/r$ is that of the radial equations.

Density, at fixed velocities. Each modulus is proportional to the density at fixed velocities, so $\rho K'_\rho = \rho K_\rho + \sum_X X K_X$: for isotropy $\rho K_\rho + \kappa K_\kappa + \mu K_\mu$ (Liu & Tromp, 2008, eq. 68), for transverse isotropy $\rho K_\rho + A K_A + C K_C + F K_F + L K_L + N K_N$. The velocity kernels follow as usual.

Source. For a moment tensor $\mathbf{M}_s$ at $\mathbf{x}_s$, stepped in time, $\langle \mathbf{s}_k^\circ, \delta\mathbf{f}_k\rangle = \delta\mathbf{M}_s : \boldsymbol{\epsilon}_k^\circ(\mathbf{x}_s)/(i\omega_k)$, giving the source kernels by reciprocity at no extra cost.

6 The adjoint potential

6.1 Strong form

Where $\mathbf{M}$ and $\mathbf{m}$ are smooth, (15) is $-(4\pi G)^{-1}\nabla^2\lambda = \nabla\cdot\nabla\cdot\mathbf{M} - \nabla\cdot\mathbf{m}$, and λ is harmonic outside V and decays at infinity. The source contains double layers where $\mathbf{M}_{nn}$ jumps (as ρ does at discontinuities), so λ is itself discontinuous there, with $[\lambda]^+_- = -4\pi G\,[\mathbf{M}_{nn}]^+_-$; the weak form handles this without comment, and in practice λ is computed from the integral representation below rather than from the strong form.

6.2 Integral representation and comparison with Eaton et al. (2026)

Taking $\psi = -G/|\mathbf{x} - \mathbf{x}_0|$ in (15), for which $\nabla^2\psi = 4\pi G\delta(\mathbf{x} - \mathbf{x}_0)$, gives $\lambda(\mathbf{x}_0) = -\int_V (\mathbf{M} : \nabla\nabla\psi + \mathbf{m}\cdot\nabla\psi)\,\mathrm{d}V$. Now $\nabla\psi = G\,\mathbf{R}/R^3$ with $\mathbf{R} = \mathbf{x} - \mathbf{x}_0$, and

$$\nabla\nabla\psi = G\,\Pi + \frac{4\pi G}{3}\mathbf{I}\,\delta(\mathbf{x} - \mathbf{x}_0), \qquad \Pi = \frac{\mathbf{I}}{R^3} - \frac{3\mathbf{R}\otimes\mathbf{R}}{R^5}, \tag{22}$$

the integral over Π being a principal value; the local term is needed for $\nabla^2\psi = 4\pi G\delta$, since Π is traceless. Hence

$$\lambda(\mathbf{x}_0) = -G\int_V \left[\mathbf{M} : \Pi + \mathbf{m}\cdot\frac{\mathbf{x}-\mathbf{x}_0}{|\mathbf{x}-\mathbf{x}_0|^3}\right] \mathrm{d}V - \frac{4\pi G}{3}\operatorname{tr}\mathbf{M}(\mathbf{x}_0). \tag{23}$$

With $\mathbf{M} \leftrightarrow \tilde{\mathbf{K}}_\rho$ and $\mathbf{m} \leftrightarrow \tilde{\mathbf{k}}_\rho$ (Eaton et al., 2026, eqs 23–24), the first term is the non-local part of their density kernel (their eq. 29 is for relative perturbations, and so carries an extra factor ρ), so the two results agree, given one remark: their eq. A4 writes $\nabla\nabla\delta\Phi = G\int\delta\rho\,\Pi$ without the local term, which is needed for $\nabla^2\delta\Phi = 4\pi G\delta\rho$. Their numerical route, which takes derivatives of solutions of Poisson's equation, includes it automatically. Here a single scalar Poisson problem replaces their nine.

6.3 In spherical harmonics

For a spherically symmetric background, expanding ψ in (15) in spherical harmonics separates the problem into one radial problem for each (l, m), whose sources are the spin components of $\mathbf{M}$ and $\mathbf{m}$, which are what the generalised spherical harmonics of GSHTrans provide. With the Green's function $-\frac{4\pi G}{2l+1}r_<^l/r_>^{l+1}$ these are radial quadratures.

6.4 One-dimensional perturbations

For $\delta\rho(r)$ the perturbation of the equilibrium potential is radial and explicit, $\delta\Phi' = 4\pi G\,\delta m(r)/r^2$ with $\delta m(r) = \int_0^r \delta\rho\, r'^2\,\mathrm{d}r'$, and $\delta\Phi'' = 4\pi G\delta\rho - 2\delta\Phi'/r$, so λ need not be formed. With $\nabla\nabla\delta\Phi = \delta\Phi''\hat{\mathbf{r}}\hat{\mathbf{r}} + (\delta\Phi'/r)(\mathbf{I} - \hat{\mathbf{r}}\hat{\mathbf{r}})$ and the angular integrals

$$\bar{M}_{rr}(r) = \oint \mathbf{M}_{rr}\,\mathrm{d}\Omega, \quad \bar{M}_h(r) = \oint (\mathbf{M}_{\theta\theta} + \mathbf{M}_{\phi\phi})\,\mathrm{d}\Omega, \quad \bar{m}_r(r) = \oint \mathbf{m}_r\,\mathrm{d}\Omega, \tag{24}$$

the gravitational part of the one-dimensional density kernel, defined by $\delta J = \int_0^a K_\rho^{1D}\,\delta\rho\, r^2\,\mathrm{d}r$, is

$$K_\rho^{1D,\Phi}(r) = -4\pi G\left[\bar{M}_{rr}(r) + \int_r^a \left(\frac{\bar{M}_h - 2\bar{M}_{rr}}{r'} + \bar{m}_r\right)\mathrm{d}r'\right], \tag{25}$$

non-local only through an integral over the radii above. The local part is the angular integral of the bracket in (20). YSpec computes g from ρ, so finite differences in the density test this kernel completely, gravitational part included.

7 Laterally varying density

A background in hydrostatic equilibrium cannot remain so under a laterally varying $\delta\rho$: the initial stress acquires a deviatoric part that the density does not determine. Like Liu & Tromp (2008) and Eaton et al. (2026), we ignore this, and perturb the hydrostatic equations as written. Two consequences should be stated. First, A_g is symmetric only for a hydrostatic background (Lemma 1); its antisymmetric part has a first-order perturbation, proportional to $\nabla_h\delta\rho$, and the kernel depends on which of $A_g(\mathbf{s}^\circ, \mathbf{s})$ and $A_g(\mathbf{s}, \mathbf{s}^\circ)$ is perturbed. Perturbing (3), as those authors do, corresponds to (5) with the adjoint field in the first slot, which we adopt. Second, the derivation of (7) dropped surface terms because $\nabla\Phi$ is normal to the boundaries; $\nabla\delta\Phi$ is not, and on Σ_{FS}, where the tangential displacements jump, perturbing the strong form generates $\int_{\Sigma_{FS}} [\rho(s_n^\circ\mathbf{s}_h - s_n\mathbf{s}_h^\circ)]_-^+ \cdot \nabla_h\delta\Phi\,\mathrm{d}\Sigma$, which those authors omit and which we also omit for consistency with them. Both terms are of the order of the inconsistency itself, and neither arises for one-dimensional perturbations.

8 Topography on discontinuities

This section is an outline, to be completed. Moving a discontinuity changes only the domains of the integrals in (13). Mapping the perturbed domain onto the reference one and using that every field satisfies its equations and transmission conditions, the first variation reduces to the jump across the boundary of the normal–normal component of the energy–momentum tensor of the Lagrangian density, $\mathbb{E} = \ell\mathbf{I} - \sum_v (\nabla v)^{\mathsf{T}} \cdot \partial\ell/\partial(\nabla v)$, summed over the fields v on which ℓ depends through first derivatives, with an additional tangential term on Σ_{FS}, where the boundary may move relative to the slipping fluid. For the elastic and inertial terms, $\ell = \rho\omega^2\mathbf{s}^\circ \cdot \mathbf{s} - \nabla\mathbf{s}^\circ : \mathbf{c} : \nabla\mathbf{s}$, and $\hat{\mathbf{n}} \cdot \mathbb{E} \cdot \hat{\mathbf{n}} = \ell + (\hat{\mathbf{n}} \cdot \mathbf{T}) \cdot \partial_n\mathbf{s}^\circ + (\hat{\mathbf{n}} \cdot \mathbf{T}^\circ) \cdot \partial_n\mathbf{s}$, which corresponds, up to their sign convention and the time domain, to the kernel of Liu & Tromp (2008, eq. 60). The constraint contributes $\left[\rho\lambda - (4\pi G)^{-1} g\, \partial_n\lambda\right]^+_-$ at a spherical boundary, and the gravitational terms of A_g involve $\nabla\nabla\Phi$, so that their energy–momentum tensor needs the second-order form. For one-dimensional problems the boundary kernel is one number per discontinuity, and finite differences that move a discontinuity will check it.

9 Evaluation in YSpec

For a spherically symmetric background, YSpec gives the forward field of each source and the field of a point force at each receiver at every radius wanted, in the frame with the source (or receiver) at the pole, as coefficients of generalised spherical harmonics with $|m| \le 2$ (or 1), and their strains as the spin components $N = -2, \ldots, 2$.

- One-dimensional kernels need only angular integrals of products of the two fields, which by orthogonality are sums over degree of products of radial functions with Wigner matrix elements at the source–receiver distance; there is no spatial mesh, and $\bar{M}_{rr}$, $\bar{M}_h$ and $\bar{m}_r$ are formed the same way.
- Three-dimensional kernels need the fields on a mesh. Expansions of spin-weighted fields rotate one N at a time, so each field is rotated from its own frame to geographic coordinates with Wigner matrices (cheaply, since only a few orders are present) and synthesised on a Gauss–Legendre grid at each radius; the kernels are pointwise products summed over frequency, and λ is the solution of the radial problems of Section 6. A cross-section through source and receiver is far cheaper than a full mesh.
- Checks: finite differences of J for one-dimensional perturbations of every parameter, density with gravity included, and the angular integrals of the three-dimensional kernels against the one-dimensional ones.

One restriction: a receiver must be where YSpec can place a source, that is, in a solid.

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